

# Curriculum Vitae

Junichi Haruna

Last updated: September 2026

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| Position     | Program-Specific Researcher, group of Keisuke Fujii, Graduate School of Informatics, Kyoto University                                                                                                  |
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| Web          | <a href="https://junichiharuna.github.io/">https://junichiharuna.github.io/</a>                                                                                                                        |
| Research IDs | ORCID 0000-0002-1828-8183; arXiv <a href="https://arxiv.org/a/haruna_j_1">https://arxiv.org/a/haruna_j_1</a> ; researchmap <a href="https://researchmap.jp/jharuna">https://researchmap.jp/jharuna</a> |
| Interests    | Quantum error correction and fault-tolerant quantum computation; algebraic topology and obstruction theory; gauge theory; the renormalization group                                                    |

## Appointments

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| 2026–     | Program-Specific Researcher, group of Keisuke Fujii, Graduate School of Informatics, Kyoto University (from April 2026).                                                                    |
| 2023–2026 | Specially Appointed Researcher, Center for Quantum Information and Quantum Biology (QIQB), Osaka University (April 2023 to March 2026). Research carried out in the group of Keisuke Fujii. |
| 2018–2019 | Research and Experiment Assistant, National Institute of Informatics (July 2018 to October 2019).                                                                                           |

## Education

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| 2020–2023 | Ph.D. course, Department of Physics and Astronomy, Graduate School of Science, Kyoto University (April 2020 to March 2023). Doctor of Science conferred March 2023. Doctoral thesis: <i>Gradient Flow Exact Renormalization Group for Scalar Field Theories</i> , supervised by Masafumi Fukuma. |
| 2018–2020 | Master's course, Department of Physics and Astronomy, Graduate School of Science, Kyoto University (April 2018 to March 2020). Master of Science, March 2020. Supervised by Hikaru Kawai.                                                                                                        |
| 2014–2018 | Faculty of Science, Kyoto University (April 2014 to March 2018). Bachelor of Science, March 2018.                                                                                                                                                                                                |

## Awards and fellowships

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| 2023      | 18th Particle Physics Medal, Encouragement Prize (Particle Physics Group of Japan), FY2023. Awarded to an early-career researcher for outstanding work in theoretical particle physics. |
| 2021–2022 | JSPS Research Fellowship for Young Scientists (DC2), principal investigator.                                                                                                            |
| 2018–2019 | JASSO Category 1 graduate scholarship, full repayment exemption for outstanding achievement.                                                                                            |

## Publications

Twelve refereed papers. The full list with links is on the website.

1. J. Haruna, “Homological origin of transversal implementability of logical diagonal gates in quantum CSS codes”, PTEP **2026**, 083A03 (2026), arXiv:2602.14499.
2. J. Haruna, “Note on Logical Gates by Gauge Field Formalism of Quantum Error Correction”, PTEP **2026**, 073B12 (2026), arXiv:2511.15224.
3. J. Haruna and K. Fujii, “Hierarchical Quantum Error Correction with Hypergraph Product Code and Rotated Surface Code”, PTEP **2025**, 103A03 (2025), arXiv:2505.18592.
4. J. Haruna, K. Shimizu and M. Yamada, “Functional renormalization group analysis of the O(3) nonlinear sigma model and non-Abelian bosonization duality”, Phys. Rev. D **111**, 045012 (2025), arXiv:2405.14627.
5. J. Haruna and M. Yamada, “Gradient Flow Exact Renormalization Group for Scalar Quantum Electrodynamics”, JHEP **05**, 291 (2024), arXiv:2312.15673.
6. K. Sakamoto, H. Morisaki, J. Haruna, E. Itou, K. Fujii and K. Mitarai, “End-to-end complexity for simulating the Schwinger model on quantum computers”, Quantum **8**, 1474 (2024), arXiv:2311.17388.

7. J. Liu, J. Haruna and M. Yamada, “Nonperturbative aspects of two-dimensional  $T\bar{T}$ -deformed scalar theory from functional renormalization group”, *Phys. Rev. D* **109**, 065016 (2024), arXiv:2309.15584.
8. Y. Abe, Y. Hamada and J. Haruna, “Fixed Point Structure of Gradient Flow Exact Renormalization Group for Scalar Field Theories”, *PTEP* **2022**, 033B03 (2022), arXiv:2201.04111.
9. J. Haruna, R. Toshio and N. Nakano, “Path integral approach to universal dynamics of reservoir computers”, *Phys. Rev. E* **107**, 034306 (2023), arXiv:2112.01886.
10. J. Haruna, K. Sakai and K. Yoshida, “Burgers equation vs. large  $N$  limit in  $T\bar{T}$ -deformed  $O(N)$  vector model”, *Nucl. Phys. B* **971**, 115499 (2021), arXiv:2104.05431.
11. J. Haruna, T. Ishii, H. Kawai, K. Sakai and K. Yoshida, “Large  $N$  analysis of  $T\bar{T}$ -deformation and unavoidable negative-norm states”, *JHEP* **04**, 127 (2020), arXiv:2002.01414.
12. J. Haruna and H. Kawai, “Weak scale from Planck scale: Mass scale generation in a classically conformal two-scalar system”, *PTEP* **2020**, 033B01 (2020), arXiv:1905.05656.

## Invited talks

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| 2026 | 2026 YITP QEC Workshop: Fault-tolerant logical processing, Yukawa Institute for Theoretical Physics, Kyoto (28 September to 9 October, scheduled); QuantHEP 2026, Queen Mary University of London (July); CREST HEP-QC One-day Workshop 2026, Kyoto University (April). |
| 2025 | Quantum Error Correction Theory Workshop for Young Researchers, Nagoya University (December); Lake Biwa Workshop (Biwako-han Kenkyukai), Shiga (March).                                                                                                                 |
| 2024 | Functional Renormalization Group at Niigata 2024, Kaishi Professional University (January).                                                                                                                                                                             |
| 2023 | Functional Renormalization Group at RIKEN 2023, RIKEN Wako Campus (January).                                                                                                                                                                                            |
| 2020 | Online 2020 NTU-Kyoto High Energy Physics Workshop (December).                                                                                                                                                                                                          |
| 2019 | Recent Progress on Field and String Theory: Kyoto-NTU 2019, Kyoto University (December).                                                                                                                                                                                |

In addition, 20 contributed talks, 9 seminars (including University College London, RIKEN, YITP and Heidelberg University) and 4 posters as of September 2026. The full list is on the website.

## Lectures

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| 2026 | Lecture on quantum error correction at “New computational methods in quantum field theory 2026”, RIKEN Wako Campus (26 to 28 January). Lecture note available on the website. |
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## Academic service

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| 2026 | Local organizer, 2026 YITP QEC Workshop: Fault-tolerant logical processing, Yukawa Institute for Theoretical Physics (28 September to 9 October). |
| 2024 | Lecturer, 3rd joint career forum, Nuclear, Particle and Cosmology Young Researchers Summer School.                                                |
| 2021 | Organizing committee (head of the preparatory school), Nuclear, Particle and Cosmology Young Researchers Summer School.                           |

## Teaching

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| 2019–2020 | Six teaching assistant appointments at Kyoto University (quantum mechanics exercises, statistics, data science, and the MACS education program). |
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## Software and data

**bockstein-gate-check**: feasibility checker for transversal diagonal logical gates in CSS codes (prototype backing arXiv:2602.14499). <https://github.com/junichiharuna/bockstein-gate-check>

Data and code for “Hierarchical Quantum Error Correction with Hypergraph Product Code and Rotated Surface Code” (with K. Fujii). Zenodo, <https://doi.org/10.5281/zenodo.15660987>

**PIReservoir**: numerical code for *Phys. Rev. E* **107**, 034306. <https://github.com/junichiharuna/PIReservoir>

**GLMCC**: Python code for the GLM-based cross-correlation method to estimate neuronal connectivity (with R. Kobayashi and S. Shinomoto). <https://github.com/NII-Kobayashi/GLMCC>